

Chapter 10 Light: Reflection and Refraction

10.1 Reflection of Light

A highly polished surface, such as a mirror, reflects most of the light falling on it. The law of reflection states that the angle of incidence is equal to the angle of reflection, and that the incident ray, the reflected ray and the normal all lie in the same plane. These laws hold for every reflecting surface, plane or spherical.

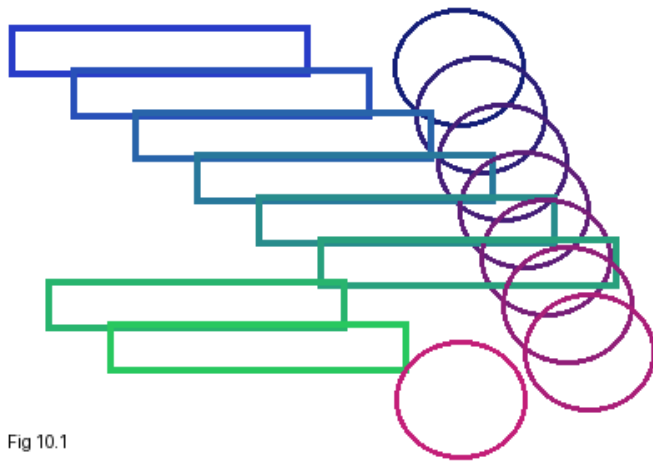


Fig 10.1

Figure 10.1 Reflection of light by a plane mirror

10.2 Spherical Mirrors

The reflecting surface of a spherical mirror may be curved inwards or outwards. A spherical mirror whose reflecting surface is curved inwards is called a concave mirror, and one curved outwards is called a convex mirror.

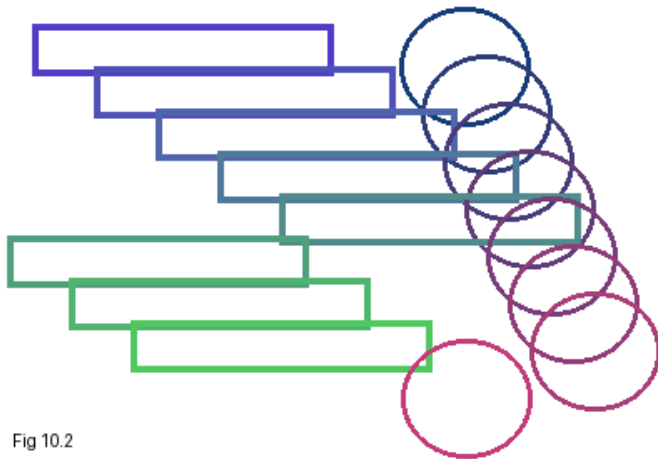


Fig 10.2

Figure 10.2 Concave and convex mirrors

10.3 Refraction of Light

Light bends while travelling from one medium into another. This bending of light is called refraction of light.

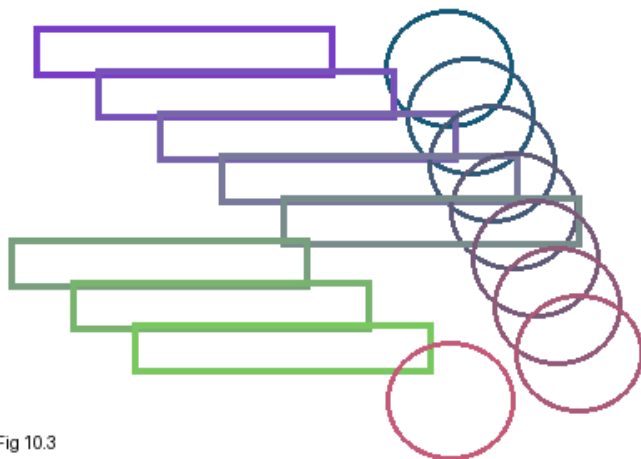


Fig 10.3

Figure 10.3 Refraction through a glass slab

Exercises

1. Define the principal focus of a concave mirror.
2. The radius of curvature of a spherical mirror is 20 cm. What is its focal length?
3. Name a mirror that can give an erect and enlarged image of an object.
4. Why do we prefer a convex mirror as a rear-view mirror in vehicles?
5. State the laws of refraction of light and explain the refractive index.